Peak Fit Applied Analysis

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Setup

```
## PACKAGES
library(tidyverse)
```

```
Warning: package 'tidyverse' was built under R version 4.5.2
```

```
Warning: package 'ggplot2' was built under R version 4.5.3
```

```
Warning: package 'readr' was built under R version 4.5.2
```

```
Warning: package 'forcats' was built under R version 4.5.2

Warning: package 'lubridate' was built under R version 4.5.2

-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.6
v forcats   1.0.1     v stringr   1.5.2
v ggplot2   4.0.3     v tibble    3.3.0
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.1.0
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```r
library(easystats)   # parameters, performance, modelbased, datawizard, insight, etc.
```

```
Warning: package 'easystats' was built under R version 4.5.3

# Attaching packages: easystats 0.7.6 (red = needs update)
x bayestestR  0.18.1   ✓ correlation 0.8.8
x datawizard  1.3.1    ✓ effectsize  1.0.3
x insight     1.5.3    x modelbased  0.17.0
x performance 0.18.0   x parameters  0.29.2
✓ report      0.6.4    x see         0.14.1

Restart the R-Session and update packages with `easystats::easystats_update()`.
```

```r
library(here)
```

```
Warning: package 'here' was built under R version 4.5.3

here() starts at C:/Users/19185/Downloads
```

```r
library(sandwich)    # robust (HC3) standard errors; added in case I need it
```

```
Warning: package 'sandwich' was built under R version 4.5.3
```

```r
## DATA
members <- here("peakfit_members.csv") |>
  read_csv()
```

```
Rows: 280 Columns: 10
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (3): membership_tier, acquisition_channel, home_location
dbl (7): member_id, age, tenure_months, baseline_engagement_score, personal_...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Section 4: Applied R Analysis

Data preparation

```
glimpse(members)
```

```
Rows: 280
Columns: 10
$ member_id                 <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, ~
$ age                       <dbl> 23, 25, 43, 40, 41, 20, 34, 21, 21, 24, 46,~
$ tenure_months             <dbl> 12, 18, 6, 29, 1, 16, 7, 2, 6, 18, 3, 9, 3,~
$ membership_tier           <chr> "Premium", "Basic", "Basic", "Basic", "Prem~
$ acquisition_channel       <chr> "Referral", "Referral", "Social Ad", "Socia~
$ home_location             <chr> "Suburban", "Suburban", "Suburban", "Urban"~
$ baseline_engagement_score <dbl> 77.3, 61.9, 51.3, 63.2, 61.0, 55.2, 46.2, 6~
$ personal_training_sessions <dbl> 3, 0, 0, 0, 0, 4, 0, 4, 0, 3, 2, 2, 1, 4, 1~
$ monthly_spend             <dbl> 145.25, 75.88, 65.63, 79.56, 109.99, 90.80,~
$ churned_year1             <dbl> 0, 0, 0, 1, 0, 0, 0, 0, 1, 1, 1, 1, 0, 0, 0~
```

```
# check for missing values
members |>
  summarize(across(everything(), \(x) sum(is.na(x)))) |>
  display()
```

member_id	age	tenure_months	membership_tier	acquisition_channel	home_location	baseline_engagement_score	personal_training_sessions	monthly_spend	churned_year1
0	0	0	0	0	0	0	0	0	0

```r
members <- members |>
  mutate(
    # creates reference levels: Basic tier; Walk-in channel; Suburban location
    membership_tier     = factor(membership_tier,
                                 levels = c("Basic", "Premium", "Elite")),
    acquisition_channel = factor(acquisition_channel,
                                 levels = c("Walk-in", "Referral", "Social Ad")),
    home_location       = factor(home_location,
                                 levels = c("Suburban", "Urban")),
    # outcome for RQ2: "active" is reference so model predicts the odds of churning
    churned_year1       = factor(churned_year1, levels = c(0, 1),
                                 labels = c("active", "churned")),
    # centers personal training so the intercept is for an average-usage member
    pt_c = personal_training_sessions - mean(personal_training_sessions)
  )

#groups by membership tier
members |>
  group_by(membership_tier) |>
  describe_distribution(select = c("monthly_spend", "personal_training_sessions")) |>
  display()
```

membership_tier	Variable	Mean	SD	IQR	Range	Skewness	Kurtosis	n	n_Missing
Basic	monthly_spend	70.46	14.03	19.30	(38.16, 119.27)	0.53	0.65	144	0
Basic	personal_training_sessions	0.94	0.96	2.00	(0.00, 4.00)	0.74	-0.22	144	0
Premium	monthly_spend	126.20	19.38	21.51	(71.88, 205.62)	0.44	2.58	92	0
Premium	personal_training_sessions	2.29	1.69	2.00	(0.00, 7.00)	0.80	0.07	92	0
Elite	monthly_spend	206.94	29.39	32.72	(146.40, 282.48)	0.54	0.46	44	0
Elite	personal_training_sessions	3.41	1.92	2.00	(0.00, 9.00)	0.87	0.88	44	0

RQ1: Spending, personal training, and membership tier

1. Analytic approach

RESPONSE:

I will use multiple linear regression (OLS) with `monthly_spend` as the outcome, because spending is continuous and roughly symmetric. To test whether the personal-training relationship differs across tiers, I will include a personal-training × membership-tier interaction, which is a moderation test: it asks whether the slope of spending on training sessions is the same in all three tiers. The model will also adjust for the other member characteristics (age, tenure, home location, acquisition channel, and early engagement score), so each coefficient reflects the association holding the others constant. I will build the model hierarchically (covariates, then tier, then personal training, then the interaction) and compare steps with Wald F-tests, so I can see whether each addition improves fit. Personal training sessions are mean-centered, so the personal-training main effect is the slope for the reference tier (Basic) at average values of the other predictors.

2. Model code

```
# Hierarchical build of my model
rq1_m0 <- lm(monthly_spend ~ 1, data = members)

rq1_m1 <- lm(monthly_spend ~ age + tenure_months + home_location +
               acquisition_channel + baseline_engagement_score,
             data = members)
#adding another factor each time
rq1_m2 <- update(rq1_m1, . ~ . + membership_tier)

rq1_m3 <- update(rq1_m2, . ~ . + pt_c)

rq1_m4 <- update(rq1_m3, . ~ . + pt_c:membership_tier)   # final model
#wald test
rq1_wald <- test_wald(rq1_m0, rq1_m1, rq1_m2, rq1_m3, rq1_m4)
display(rq1_wald, digits = 3)
```

Name	Model	df	df_diff	F	p
rq1_m0	lm	279			
rq1_m1	lm	273	6	44.209	< .001
rq1_m2	lm	271	2	2096.934	< .001
rq1_m3	lm	270	1	247.832	< .001
rq1_m4	lm	268	2	25.692	< .001

Models were detected as nested (in terms of fixed parameters) and are compared in sequential order.

```
# Final model parameters
rq1_mpm <- model_parameters(rq1_m4, ci = 0.95)
display(rq1_mpm, digits = 3)
```

Parameter	Coefficient	SE	95% CI	t(268)	p
(Intercept)	59.847	4.657	(50.678, 69.015)	12.851	< .001
age	-0.083	0.080	(-0.240, 0.074)	-1.045	0.297
tenure months	0.370	0.050	(0.272, 0.469)	7.419	< .001
home location (Urban)	1.041	1.523	(-1.958, 4.041)	0.684	0.495
acquisition channel (Referral)	-2.748	1.998	(-6.682, 1.185)	-1.376	0.170
acquisition channel (Social Ad)	-5.426	1.941	(-9.248, -1.603)	-2.795	0.006
baseline engagement score	0.219	0.048	(0.125, 0.314)	4.590	< .001
membership tier (Premium)	48.085	1.933	(44.280, 51.889)	24.882	< .001
membership tier (Elite)	110.683	2.810	(105.151, 116.215)	39.392	< .001
pt c	2.983	1.095	(0.826, 5.139)	2.723	0.007
membership tier (Premium) × pt c	4.761	1.337	(2.129, 7.393)	3.561	< .001
membership tier (Elite) × pt c	10.624	1.497	(7.677, 13.572)	7.098	< .001

```
# Overall fit
rq1_mpf <- model_performance(rq1_m4)
display(rq1_mpf, digits = 3)
```

AIC	AICc	BIC	R2	R2 (adj.)	RMSE	Sigma
2214.4	2215.8	2261.7	0.947	0.944	12.048	12.315

```
# Direct test of the interaction (final model vs. same model without it)
rq1_int_test <- test_wald(rq1_m3, rq1_m4)
display(rq1_int_test, digits = 3)
```

Name	Model	df	df_diff	F	p
rq1_m3	lm	270			
rq1_m4	lm	268	2	25.692	< .001

Models were detected as nested (in terms of fixed parameters) and are compared in sequential order.

```
# Simple slope of personal training within each tier (dollars per extra session)
rq1_slopes <- estimate_slopes(rq1_m4, trend = "pt_c", by = "membership_tier")
display(rq1_slopes, digits = 2)
```

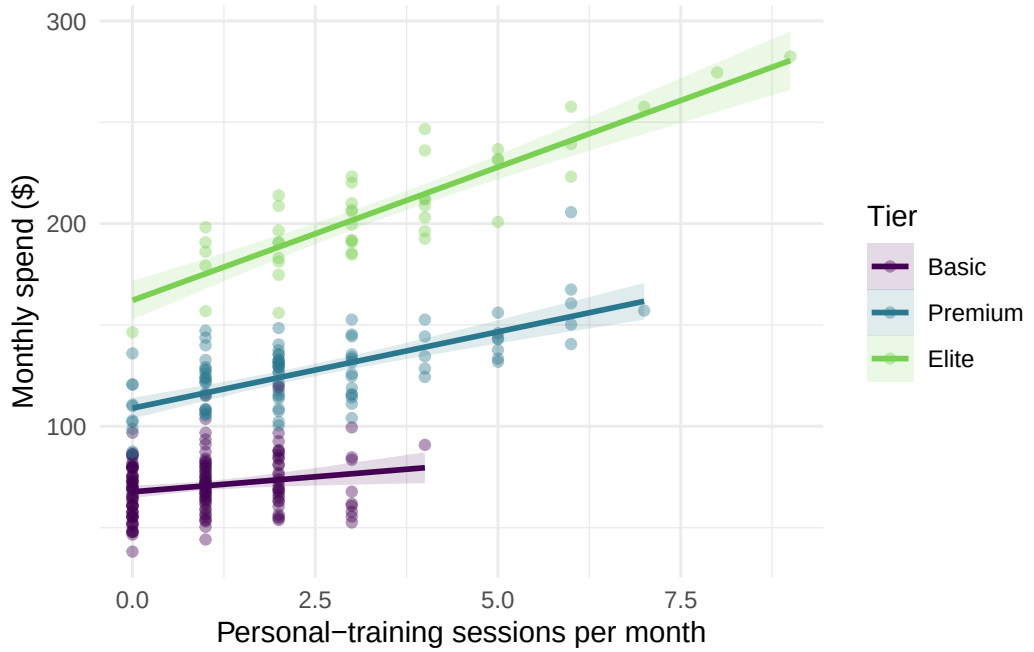
Table 1: Estimated Marginal Effects

membership_tier	Slope	SE	95% CI	t(268)	p
Basic	2.98	1.10	(0.83, 5.14)	2.72	0.007
Premium	7.74	0.76	(6.24, 9.25)	10.12	< .001
Elite	13.61	0.99	(11.65, 15.56)	13.69	< .001

Marginal effects estimated for pt_c; Type of slope was dY/dX

```
# Gives Raw data with a separate line for each tier (unadjusted)
ggplot(members, aes(x = personal_training_sessions, y = monthly_spend,
                    colour = membership_tier, fill = membership_tier)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = TRUE, alpha = 0.15) +
  scale_colour_viridis_d(end = 0.8) +
  scale_fill_viridis_d(end = 0.8) +
  labs(x = "Personal-training sessions per month", y = "Monthly spend ($)",
       colour = "Tier", fill = "Tier") +
  theme_minimal()
```

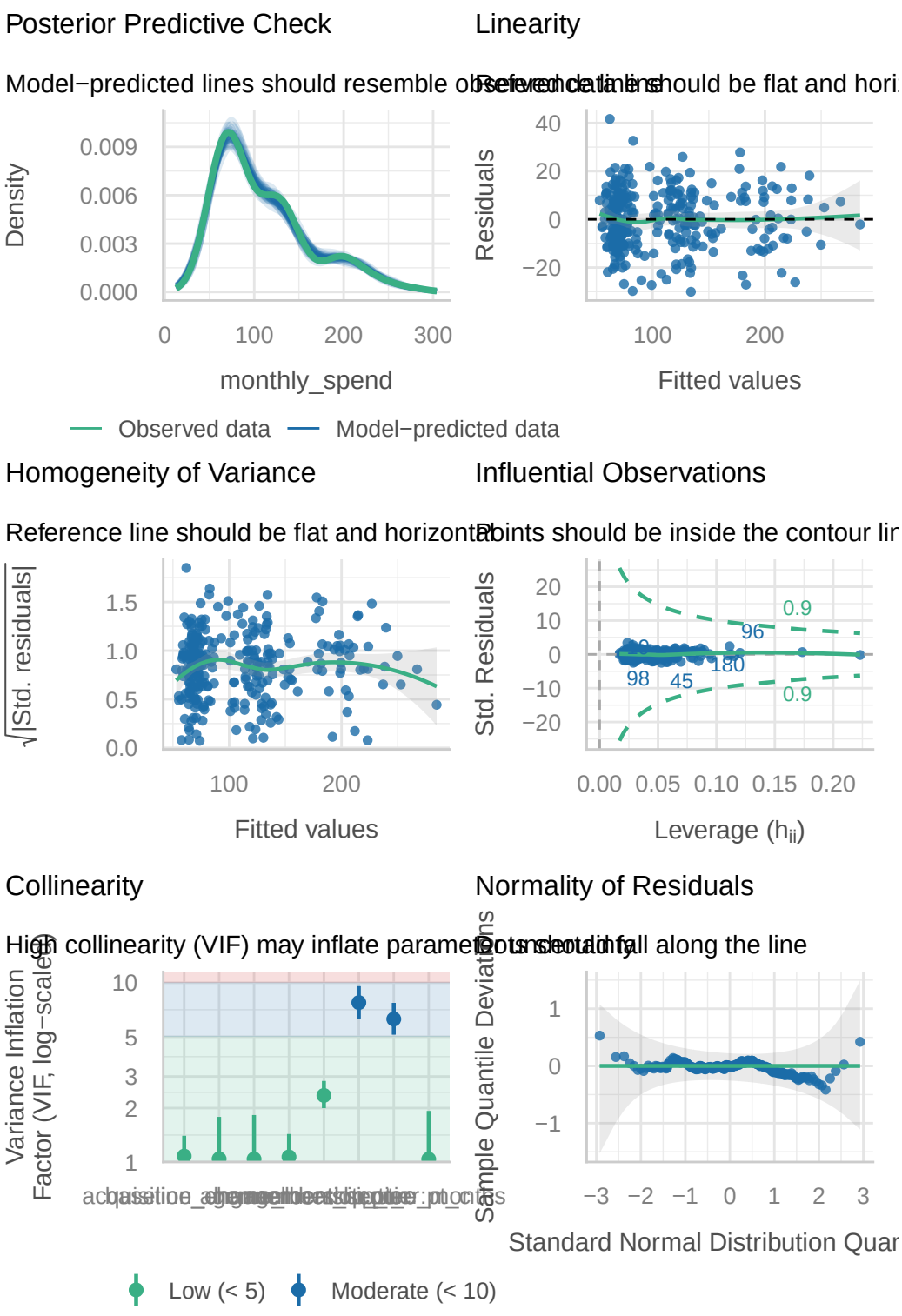
```
`geom_smooth()` using formula = 'y ~ x'
```



RESPONSE (code summary): I built the model in steps (covariates, then tier, then personal training, then the interaction) and compared each step with a Wald F-test. Personal training is mean-centered, and the reference groups are Basic (tier), Walk-in (channel), and Suburban (location). The key code is in the exam submission.

3. Assumption check

```
# makes diagnostic plots (linearity, homogeneity of variance, influence, normality of residuals
check_model(rq1_m4)
```




```
# Formal test of the assumption I am reporting (heteroskedasticity)
check_heteroscedasticity(rq1_m4)
```

```
OK: Error variance appears to be homoscedastic (p = 0.684).
```

```
# Supporting check: normality of residuals
check_normality(rq1_m4)
```

```
OK: residuals appear as normally distributed (p = 0.234).
```

```
# Collinearity is checked on the model WITHOUT the interaction, because
# an interaction term is built from its components and inflates VIF by design.
# Therefore, collinearity is just checked on m3.
check_collinearity(rq1_m3) |>
  display(digits = 2)
```

Term	VIF	VIF_CI_low	VIF_CI_high	SE_factor	Tolerance	Tolerance_CI_low	Tolerance_CI_high
age	1.03	1.00	2.10	1.02	0.97	0.48	1.00
tenure_months	1.03	1.00	2.12	1.02	0.97	0.47	1.00
home_location	1.03	1.00	3.20	1.01	0.97	0.31	1.00
acquisition_channel	1.06	1.01	1.51	1.01	0.95	0.66	0.99
baseline_engagement_score	1.04	1.00	1.95	1.02	0.96	0.51	1.00
membership_tier	1.48	1.30	1.75	1.10	0.68	0.57	0.77
pt_c	1.45	1.28	1.72	1.20	0.69	0.58	0.78

```
# Same model, heteroskedasticity-consistent (HC3) standard errors:
model_parameters(rq1_m4, vcov = "HC3") |>
  display(digits = 3)
```

RESPONSE:

The assumption I checked was constant variance in the residuals (homoscedasticity),because if the spread changes across fitted values, the standard errors and p-values can't be trusted. The test came back fine (p = 0.684), and the residual plots show an even band with no funnel shape. The residuals also looked normal (p = 0.234), no observation had unusual influence, and collinearity wasn't a problem (the largest VIF in the main-effects model was 1.48, for tier). The higher VIFs in the full-model plot come from the interaction term overlapping with its own components, which is expected. Since nothing was wrong, no fix was needed and I kept the ordinary OLS results. As a precaution I also re-ran the model with HC3 robust standard errors, and the conclusions didn't change: the same terms were significant (Social Ad p = 0.009, personal training p = .009, and both interactions p < 0.001) and the same terms were not (Referral p = 0.173, age p = 0.301, location p = 0.511).

4. Key results

RESPONSE: Yes, the relationship between personal training and spending differs by tier. Adding the personal training × tier interaction significantly improved the model, $F(2, 268) = 25.69$, $p < .001$. Earlier steps also helped: the covariates beat the intercept-only model, $F(6, 273) = 44.21$, $p < .001$; tier added a lot, $F(2, 271) = 2096.93$, $p < .001$; and personal training added more, $F(1, 270) = 247.83$, $p < .001$. The full model explains about 95% of the variation in monthly spend ($R^2 = .947$, adjusted $R^2 = .944$). Training is linked to higher spending in every tier, but the payoff grows with the tier. Each extra personal-training session per month goes with $2.98 more spending for Basic members (95% CI [0.83, 5.14], $p = .007$), 7.74 more for Premium (95% CI [6.24, 9.25], $p < .001$), and $13.61 more for Elite (95% CI [11.65, 15.56], $p < .001$). Compared with Basic, the slope is $4.76 steeper for Premium (95% CI [2.13, 7.39], $p < .001$) and 10.62 steeper for Elite (95% CI [7.68, 13.57], $p < .001$). The scatterplot shows the same thing: an almost flat line for Basic and much steeper lines for Premium and Elite. Other characteristics also mattered. Members who had been around longer spent more ($b = 0.37$ more dollars per month of tenure, $p < .001$), and so did members with higher early engagement ($b = 0.22$ per point, $p < .001$). Social Ad members spent about 5.43 less per month than Walk-ins (95% CI [-9.25, -1.60], $p = .006$), while Referral members weren't significantly different from Walk-ins ($p = .170$). Age ($p = .297$) and home location ($p = .495$) weren't significant. Tier itself had a huge effect: at average training use, Premium members spent about $48 more than Basic members and Elite members about $111 more (both $p < .001$).

5. Plain-language summary and recommendation

RESPONSE: Members who do more personal training spend more, but how much more depends on their membership level. Each extra session a month is tied to about $3 more spending for a Basic member, about $8 for a Premium member, and about $14 for an Elite member. That holds after accounting for age, how long they've been a member, where they live, how they joined, and how engaged they were early on. So personal training is most valuable among Premium and Elite members. Members who've been with PeakFit longer and those who were more engaged in their first two weeks also spend more, and members who joined through a social ad spend about $5 less per month than walk-ins.

Recommendation: Promote personal training to Premium and Elite members, for example with a free or discounted first session or a training package offered at sign-up or upgrade, since that's where each extra session is linked to the biggest increase in spending. This only shows an association, so it doesn't prove that training causes the extra spending (big spenders may just buy more of everything). I'd pilot the offer at a few locations and track results before rolling it out chain-wide.

RQ2: Predicting first-year cancellation

6. Analytic approach

RESPONSE:

I'll use binary logistic regression, because the outcome is yes/no (cancelled within the first year or not). It models the odds of cancelling, so the coefficients convert directly into odds ratios, which is what the question asks for across acquisition channels. Walk-in is the reference group, so Referral and Social Ad are each compared with Walk-in. I'll also include age, tenure, home location, membership tier, and early engagement score. I left out personal-training sessions and monthly spend on purpose, since they're measured during membership and depend partly on how long someone stayed, so using them to predict cancellation would be circular. The early engagement score only covers the first 14 days, so it's fair to use. I included tenure because recent joiners have had less time to cancel. There are only 33 cancellations, so I kept the model small.

7. Model code

```
# gives quick look at the outcome by channel (row percentages)
members |>
  data_tabulate(select = "acquisition_channel", by = "churned_year1",
                proportions = "row") |>
  display(digits = 1)
```

acquisition_channel	active	churned	(NA)	Total
Walk-in	52 (77.6%)	15 (22.4%)	0 (0.0%)	67
Referral	88 (91.7%)	8 (8.3%)	0 (0.0%)	96
Social Ad	107 (91.5%)	10 (8.5%)	0 (0.0%)	117
(NA)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0
Total	247	33	0	280

```
# done by umber of cancellations (events) relative to number of predictors
members |>
  count(churned_year1) |>
  display()
```

churned_year1	n
active	247
churned	33

```r
# i got the structure of this from the lectures
rq2_model <- glm(
  churned_year1 ~ acquisition_channel + age + tenure_months + membership_tier +
    home_location + baseline_engagement_score,
  family = binomial(),
  data = members
)

# Model vs. intercept-only (null) model
rq2_lrt <- test_lrt(rq2_model)
```

```
Only one model was provided, however, at least two are required for
  comparison.
  Fitting a null-model as reference now.
```

```r
display(rq2_lrt, digits = 3)
```

Table 2: Likelihood-Ratio-Test (LRT) for Model Comparison (ML-estimator)

Name	Model	df	df_diff	Criterion	Chi2	p
Null model	glm	1		203.07		
Full model	glm	9	8	169.88	33.20	< .001

```r
# Model fit
rq2_mpf <- model_performance(rq2_model)
display(rq2_mpf, digits = 3)
```

AIC	AICc	BIC	Tjur's R2	RMSE	Sigma	Log_loss	Score_log	Score_spherical	PCP
187.9	188.5	220.6	0.141	0.298	1	0.303	-4.264	0.038	0.821

```r
# Log-odds
rq2_mpm <- model_parameters(rq2_model)
display(rq2_mpm, digits = 3)
```

Parameter	Log-Odds	SE	95% CI	z	p
(Intercept)	3.237	1.214	(0.899, 5.692)	2.666	0.008
acquisition channel (Referral)	-1.676	0.534	(-2.783, -0.669)	-3.140	0.002
acquisition channel (Social Ad)	-1.770	0.498	(-2.785, -0.816)	-3.550	< .001
age	-0.024	0.022	(-0.069, 0.019)	-1.090	0.276

Parameter	Log-Odds	SE	95% CI	z	p
tenure months	-0.033	0.018	(-0.072, -8.275e-05)	-1.803	0.071
membership tier (Premium)	-0.413	0.458	(-1.351, 0.460)	-0.903	0.366
membership tier (Elite)	-1.255	0.696	(-2.818, -0.017)	-1.804	0.071
home location (Urban)	0.282	0.407	(-0.511, 1.094)	0.694	0.488
baseline engagement score	-0.049	0.014	(-0.077, -0.023)	-3.609	< .001

```
# Odds ratios
rq2_or <- model_parameters(rq2_model, exponentiate = TRUE)
display(rq2_or, digits = 3)
```

Parameter	Odds Ratio	SE	95% CI	z	p
(Intercept)	25.453	30.905	(2.457, 296.617)	2.666	0.008
acquisition channel (Referral)	0.187	0.100	(0.062, 0.512)	-3.140	0.002
acquisition channel (Social Ad)	0.170	0.085	(0.062, 0.442)	-3.550	< .001
age	0.976	0.022	(0.933, 1.019)	-1.090	0.276
tenure months	0.968	0.018	(0.931, 1.000)	-1.803	0.071
membership tier (Premium)	0.661	0.303	(0.259, 1.585)	-0.903	0.366
membership tier (Elite)	0.285	0.198	(0.060, 0.983)	-1.804	0.071
home location (Urban)	1.326	0.539	(0.600, 2.987)	0.694	0.488
baseline engagement score	0.952	0.013	(0.926, 0.977)	-3.609	< .001

```
# Predicted probability of cancelling by channel (other predictors at typical values)
estimate_means(rq2_model, by = "acquisition_channel") |>
  display(digits = 3)
```

Table 3: Estimated Marginal Means

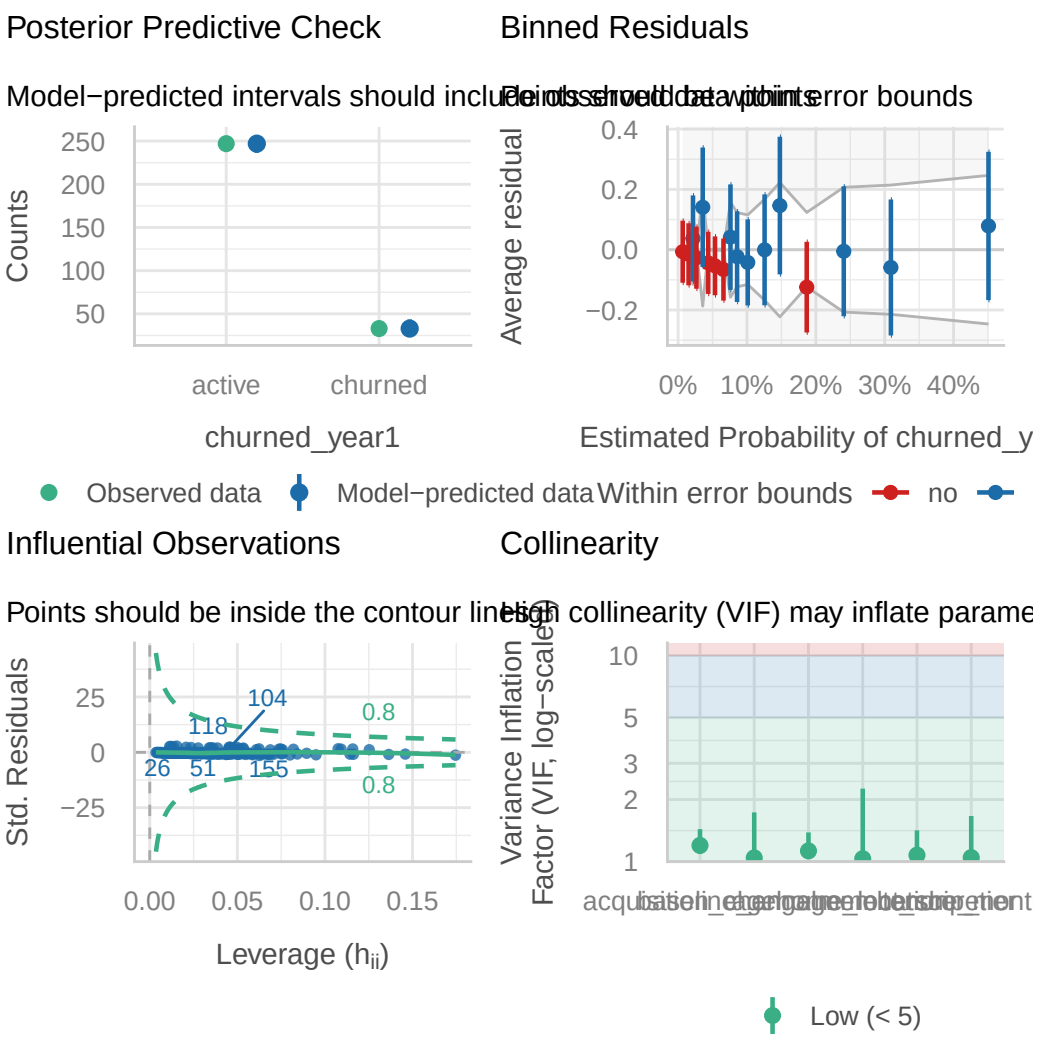
acquisition_channel	Probability	95% CI
Walk-in	0.192	(0.108, 0.319)
Referral	0.043	(0.017, 0.104)
Social Ad	0.039	(0.017, 0.089)

Variable predicted: churned_year1; Predictors modulated: acquisition_channel; Predictors averaged: age (34), tenure_months (17), membership_tier, home_location, baseline_engagement_score (58); Predictions are on the response-scale.

```
# Collinearity
check_collinearity(rq2_model) |>
  display(digits = 2)
```

Term	VIF	VIF_CI_low	VIF_CI_high	SE_factor	Tolerance	Tolerance_CI_low	Tolerance_CI_high
acquisition_channel	1.20	1.09	1.44	1.09	0.83	0.70	0.92
age	1.04	1.00	1.73	1.02	0.96	0.58	1.00
tenure_months	1.05	1.00	1.66	1.02	0.96	0.60	1.00
membership_tier	1.08	1.01	1.42	1.04	0.93	0.70	0.99
home_location	1.03	1.00	2.26	1.02	0.97	0.44	1.00
baseline_engagement_score	1.13	1.04	1.39	1.06	0.89	0.72	0.96

```
# Diagnostic plots
check_model(rq2_model)
```




```
# Linearity of the logit for each continuous predictor (meaning x * log(x) test)
rq2_logit_test <- glm(
  churned_year1 ~ acquisition_channel + age + tenure_months + membership_tier +
    home_location + baseline_engagement_score +
```

```
    age:log(age) +
    tenure_months:log(tenure_months) +
    baseline_engagement_score:log(baseline_engagement_score),
  family = binomial(),
  data = members
)
model_parameters(rq2_logit_test) |>
  display(digits = 3)
```

Parameter	Log-Odds	SE	95% CI	z	p
(Intercept)	2.800	4.990	(-7.473, 12.317)	0.561	0.575
acquisition channel (Referral)	-1.692	0.542	(-2.819, -0.672)	-3.123	0.002
acquisition channel (Social Ad)	-1.791	0.503	(-2.815, -0.830)	-3.563	< .001
age	0.036	0.637	(-1.150, 1.378)	0.057	0.954
tenure months	-0.106	0.137	(-0.367, 0.178)	-0.775	0.438
membership tier (Premium)	-0.394	0.464	(-1.345, 0.493)	-0.850	0.396
membership tier (Elite)	-1.262	0.700	(-2.832, -0.014)	-1.803	0.071
home location (Urban)	0.276	0.410	(-0.524, 1.095)	0.673	0.501
baseline engagement score	-0.012	0.205	(-0.398, 0.434)	-0.057	0.954
age × log(age)	-0.014	0.142	(-0.313, 0.250)	-0.098	0.922
tenure months × log(tenure months)	0.019	0.035	(-0.055, 0.084)	0.544	0.586
baseline engagement score × log(baseline engagement score)	-0.008	0.043	(-0.101, 0.072)	-0.186	0.852

RESPONSE (code summary): Walk-in is the reference channel, and the outcome is coded so the model predicts the odds of cancelling. Key model is in the exam submission. For assumptions, all VIFs were 1.20 or lower, and none of the predictor × log(predictor) terms was significant (age p = .922, tenure p = .586, engagement p = .852), so the linearity-of-the-logit assumption looks fine. The diagnostic plots showed no influential cases. One caution is that 33 cancellations is a small number for eight predictors, so the estimates are less precise than they would be with more events.

8. Overall model fit

RESPONSE:

Yes. The full model fit significantly better than the intercept-only model, likelihood-ratio $\chi^2(8) = 33.20$, $p < .001$ (Tjur's $R^2 = .141$, AIC = 187.9). The predictors together separate members who cancelled from those who stayed better than chance, though a Tjur's R^2 of .14 means a good deal of cancellation is still unexplained.

9. Acquisition channel and other predictors

RESPONSE: Acquisition channel matters a lot. Compared with Walk-in members, the odds of cancelling were about 81% lower for Referral members (OR = 0.19, 95% CI [0.06, 0.51], p = .002) and about 83% lower for Social Ad members (OR = 0.17, 95% CI [0.06, 0.44], p < .001), holding the other predictors constant. Flipped around, Walk-ins had roughly 5 to 6 times the odds of cancelling compared with either group. Referral and Social Ad looked about the same as each other. The raw rates tell the same story: 22.4% of Walk-ins cancelled, versus 8.3% of Referral and 8.5% of Social Ad members. After adjusting for the other predictors, the model estimates 19.2% for Walk-in, 4.3% for Referral, and 3.9% for Social Ad. The only other significant predictor was early engagement score. Each additional point was linked to 5% lower odds of cancelling (OR = 0.95, 95% CI [0.93, 0.98], p < .001), which works out to about 39% lower odds for every 10 points. Tenure (OR = 0.97, p = .071) and Elite tier (OR = 0.29, p = .071) leaned toward lower odds of cancelling but didn't reach significance. Age (p = .276), Premium tier (p = .366), and home location (p = .488) weren't significant.

AI Disclosure: Response

I only used it to see if I had everything. Everything else, I used the stuff given by the professor to finish the exam.